Derivatives Securities - Lecture 7 Trading Strategies Involving Options

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Trading strategies

A strategy is a set of positions (a portfolio) to achieve a particular risk/return profile.

We will consider following trading strategies:

- Zero-coupon bond plus option (principal-protected note)
- Stock plus option
- Two or more options of the same type (a spread)
    - Bull spread, Bear spread, Butterfly spread
- Two or more options of different types (a combination)
    - Box, Straddle, Strangle

Assumptions for simplicity:

- All options European
- On the same underlying
- With the same expiration

Investors choose different payoff structures based on their risk tolerance and their expectations about future price movements.

Principal-protected notes

- Structured investment products that guarantee the return of the original investment at maturity while offering potential returns linked to the performance of an underlying asset.

- Allows investors to take a risky position without risking the initial principal amount invested – appeals to risk-averse investors

- If you believe that asset value will increase – buy a principal-protected note consisting of:
  - A call option with a strike price of K (usually at-the-money)
  - A zero-coupon bond with a par value of K

Position	Initial *Cost*	Terminal *Payoff*	
		$\boldsymbol{If\ S_T < K}$	$\boldsymbol{If\ S_T \geq K}$
Long call	c	0	$S_T - K$
Long bond	$K * e^{-r*T}$	K	K
Total	$\boldsymbol{c + K * e^{-r*T}}$	$\boldsymbol{K}$	$\boldsymbol{K + (S_T - K)}$

Principal-protected notes
Example

A Bank offers a $1,000 instrument (principal protected note) consisting of

1-year zero-coupon bond with principal of $1,000

1-year at-the-money call option on a stock portfolio currently worth $1,000 (i.e. K=$1,000)

Investor pays ***$1,000*** today and will get the maximum of ***$1,000*** and $\boldsymbol{S_T}$ at expiration.

Profit = max(0, S_T - $1,000)

The bank will build in a profit for itself so initial cost of strategy to the bank < $1,000

$$\boldsymbol{c + K * e^{-r*T} < \$1,000}$$

Principal-protected notes

If you believe that asset value will decrease – buy a principal-protected note consisting of:

- A put option with a strike price of K (usually at-the-money)
- A zero-coupon bond with a par value of K

Position	Initial *Cost*	Terminal *Payoff*	
		$\boldsymbol{If\ S_T < K}$	$\boldsymbol{If\ S_T \geq K}$
Long put	p	$K - S_T$	0
Long bond	$K * e^{-r*T}$	K	K
Total	$\boldsymbol{p + K * e^{-r*T}}$	$\boldsymbol{K + (K - S_T)}$	$\boldsymbol{K}$

Again, the note offered by a bank will cost more than $\boldsymbol{p + K * e^{-r*T}}$

Principal-protected notes
Example

A bank decides to create a two-year principal-protected note on a non-dividend-paying stock by offering investors a zero-coupon bond with a face value of \$100 plus a put option with K=\$100. If the bank sets the cost of the note at \$100 and interest rates are currently 8% per annum for all maturities, what is the maximum cost of the put option for the bank?

How will the answer change if interest rates were 4% instead?

Principal-protected notes' feasibility

Viability depends on:

- Level of interest rates (bond prices decrease with interest rates; higher rates= cheaper capital protection)
- Volatility of the portfolio (option prices increase with volatility)
- Level of dividends (higher dividends reduce call option values)

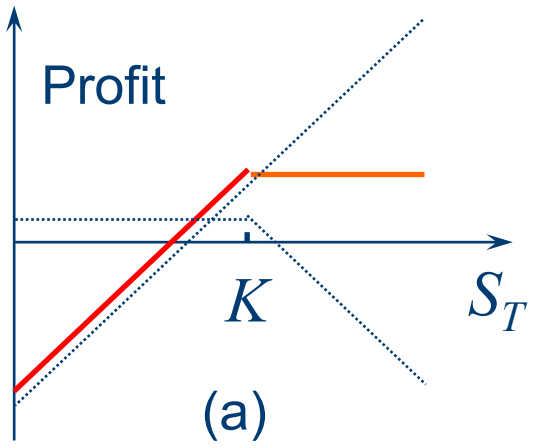
When volatility is high and interest rates are low, the bank could alter the note to make it viable:

- Out of the money strike price (lower option cost)
- Caps on investor return
- Increasing maturity

Positions in single option and the underlying stock

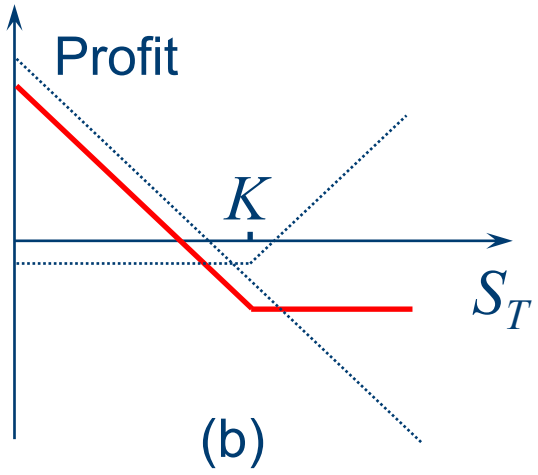
(a) Long stock & short European call (**writing a covered call**)

- Earn option premium upfront
- Downside risk remains
- Upside is limited
- Suitable if you **expect moderate rise or sideways movement**



(b) Short stock & long European call **(synthetic put)**

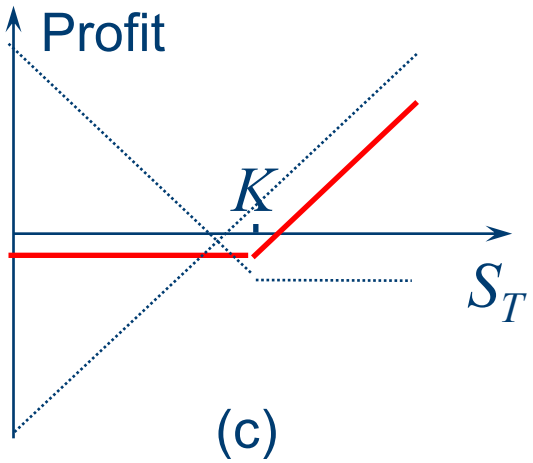
- Profit when stock declines
- Loss limited if stock rises
- Equivalent to buying a put option
- Use if you are **bearish on the stock** & want downside profit with limited upside risk



Positions in single option and the underlying stock

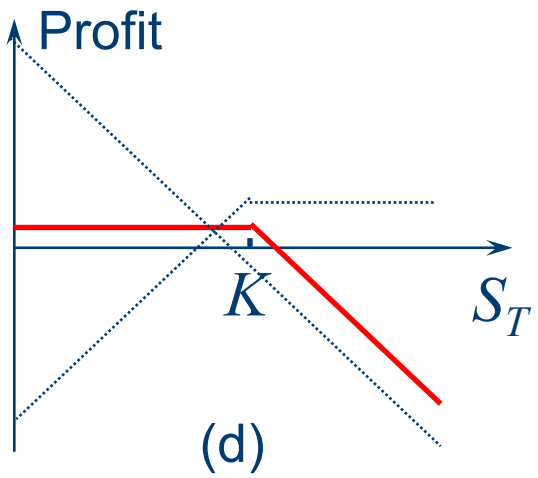
(c) Long stock & long European put (**protective put**)

- Sets a floor at the strike price
- Upside remains unlimited
- Similar to buying insurance on the stock
- Suitable if you are **bullish but want protection**



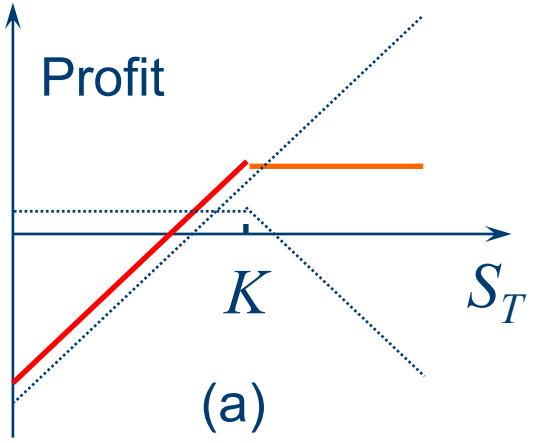
(d) Short stock & short European put **(writing a covered put)**

- Profit when stock declines
- Gains are capped once price falls below K
- Suitable if you are **moderately bearish** and willing to be assigned the stock.



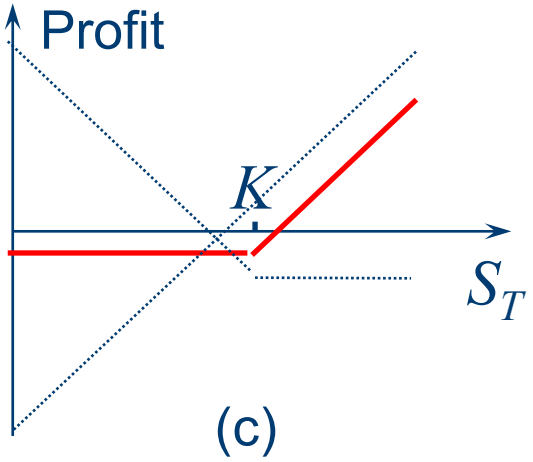
Writing a covered call
Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*	
		$\boldsymbol{If\ S_T < K}$	$\boldsymbol{If\ S_T \geq K}$
Long stock	S_0	S_T	S_T
Write call	$-c$	0	$K - S_T$
Total	$\boldsymbol{S_0 - c}$	$\boldsymbol{S_T}$	$\boldsymbol{K}$
	Total Profit:	$\boldsymbol{S_T - S_0 + c}$	$\boldsymbol{K - S_0 + c}$



A protective put
Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*	
		$\boldsymbol{If\ S_T < K}$	$\boldsymbol{If\ S_T \geq K}$
Long stock	S_0	S_T	S_T
Long put	p	$K - S_T$	0
Total	$\boldsymbol{S_0 + p}$	$\boldsymbol{K}$	$\boldsymbol{S_T}$
	Total Profit:	$\boldsymbol{K - S_0 - p}$	$\boldsymbol{S_T - S_0 - p}$

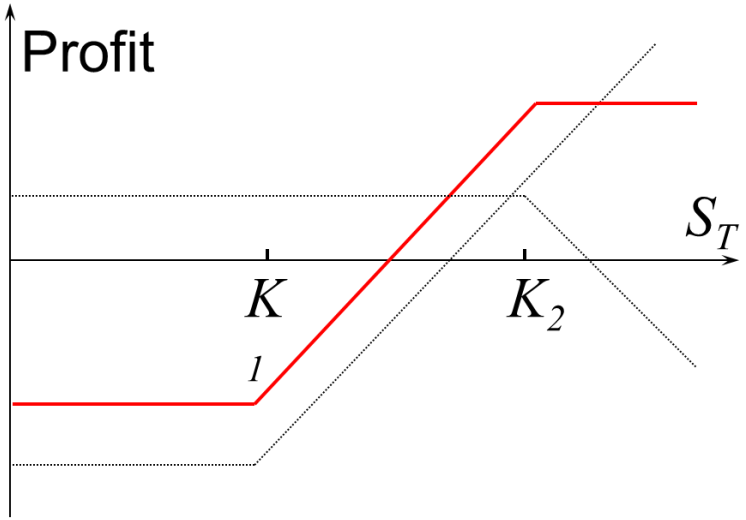


Bull Spread

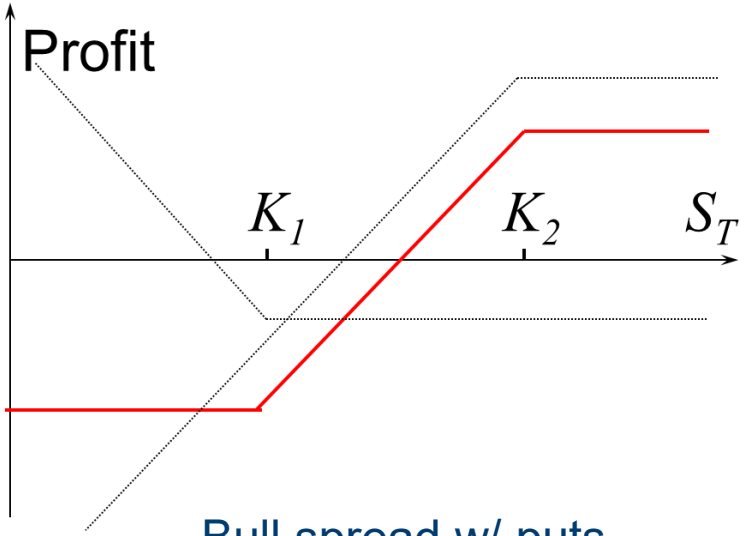
- A spread strategy with two calls (or two puts) on the same underlying and same maturity but different strike prices ($K_1 < K_2$)
- Investor is hoping spot price will <u>increase (moderately)</u>
- Limits both downside risk and upside potential
- Selling option helps finance buying another one

- With *<u>calls</u>* has initial *<u>outflow</u>*:
  - Buy call with $K_{1\ \text{(more valuable)}}$
  - Sell call with K_2

- With *<u>puts</u>* has initial *<u>inflow</u>*:
  - Buy put with K_1
  - Sell put with $K_{2\ \text{(more valuable)}}$



Bull spread w/ calls



Bull spread w/ puts

Bull Spread Using Calls
Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*		
		$If\ S_T \leq K_1$ (no exercise)	$If\ K_1 \leq S_T \leq K_2$ (one exercise)	$If\ S_T \geq K_2$ (both exercise)
Long call w/ K_1	c_1	0	$S_T - K_1$	$S_T - K_1$
Short call w/ K_2	$-c_2$	0	0	$K_2 - S_T$
Total	$c_1 - c_2$	**0**	$S_T - K_1$	$K_2 - K_1$
	Total Profit:	$-(c_1 - c_2)$	$S_T - K_1 - (c_1 - c_2)$	$K_2 - K_1 - (c_1 - c_2)$

- $S_T^{Break-even} = K_1 + c_1 - c_2$
- $Profit_T^{Minimum} = -(c_1 - c_2)$
- $Profit_T^{Maximum} = K_2 - K_1 - (c_1 - c_2)$

Bear Spread

- The reverse of the bull spread positions
- Investor is hoping spot price will <u>decrease</u>
- Limits both downside risk and upside potential

With *calls* has initial *inflow*:

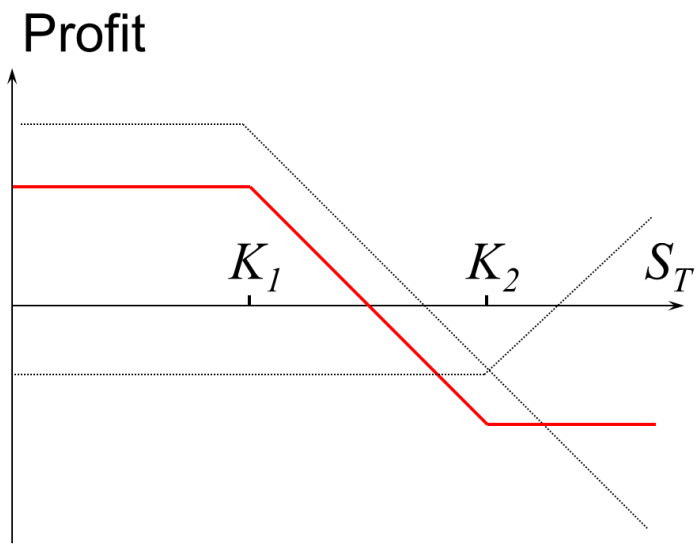
Sell call with K_1

Buy call with K_2

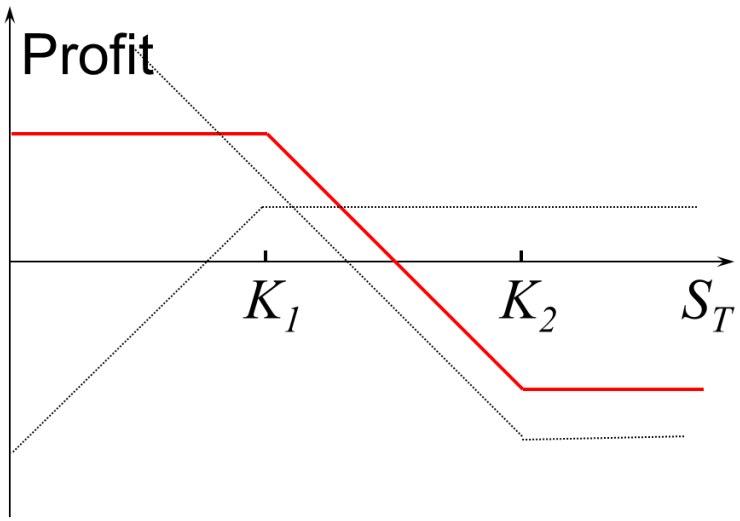
With *puts* has initial *outflow*:

Sell put with K_1

Buy put with K_2



Bear spread w/ calls



Bear spread w/ puts

Bear Spread Using Puts
Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*		
		$If\ S_T \leq K_1$ (both exercise)	$If\ K_1 \leq S_T \leq K_2$ (one exercise)	$If\ S_T \geq K_2$ (no exercise)
Short put w/ K_1	$-p_1$	$S_T - K_1$	0	0
Long put w/ K_2	p_2	$K_2 - S_T$	$K_2 - S_T$	0
Total	$\boldsymbol{p_2 - p_1}$	$\boldsymbol{K_2 - K_1}$	$\boldsymbol{K_2 - S_T}$	$\boldsymbol{0}$
	Total Profit:	$\boldsymbol{K_2 - K_1 - (p_2 - p_1)}$	$\boldsymbol{K_2 - S_T - (p_2 - p_1)}$	$\boldsymbol{-(p_2 - p_1)}$

- $\boldsymbol{S_T^{Break-even} = K_2 - (p_2 - p_1)}$
- $\boldsymbol{Profit_T^{Minimum} = -(p_2 - p_1)}$
- $\boldsymbol{Profit_T^{Maximum} = K_2 - K_1 - (p_2 - p_1)}$

Box Spread

- Combination of a bull call spread and a bear put spread:
  - Buy call with K_1
  - Sell call with K_2
  - Sell put with K_1
  - Buy put with K_2

$$\boldsymbol{Initial\ Cost} = (\boldsymbol{c_1} - \boldsymbol{c_2}) + (\boldsymbol{p_2} - \boldsymbol{p_1})$$

- The payoff of a box spread is <u>always</u> $(\boldsymbol{K_2} - \boldsymbol{K_1})$ (i.e. it's independent of S_T) – risk free
- If the payoff is certain, then the cost of a strategy should be $(\boldsymbol{K_2} - \boldsymbol{K_1}) * \boldsymbol{e^{-rT}}$
- If $\boldsymbol{Initial\ Cost} \neq (\boldsymbol{K_2} - \boldsymbol{K_1}) * \boldsymbol{e^{-rT}}$, then we have an arbitrage opportunity

Only true with European options!

*it's like lending money (synthetic lending)

*reverse box is like borrowing (synthetic loan)

*box spreads are mainly used by professional traders and institutions to exploit pricing inefficiencies or obtain synthetic financing.

Butterfly Spread

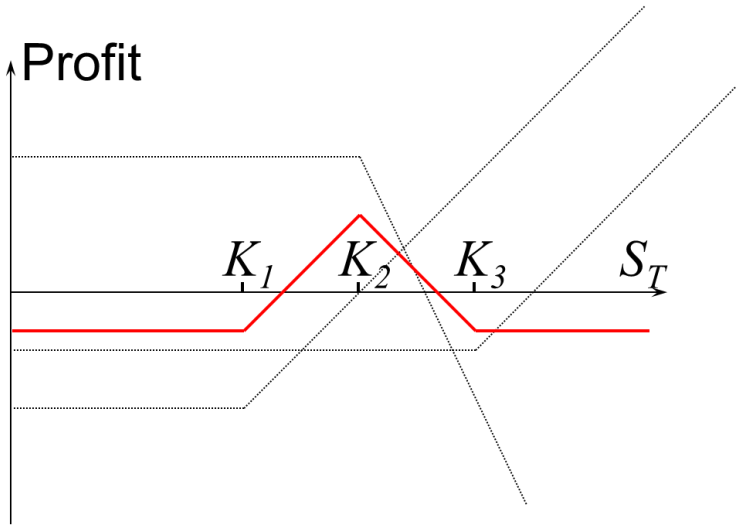
- A spread strategy with three calls (or three puts) on the same underlying and same maturity but different strike prices ($K_1 < K_2 < K_3$), where $K_2 = (K_1 + K_3)/2$
- Investor is hoping spot price will stay approximately the same (i.e. feels large moves are unlikely)
- Limits both downside risk and upside potential

With *calls* has initial *outflow*:

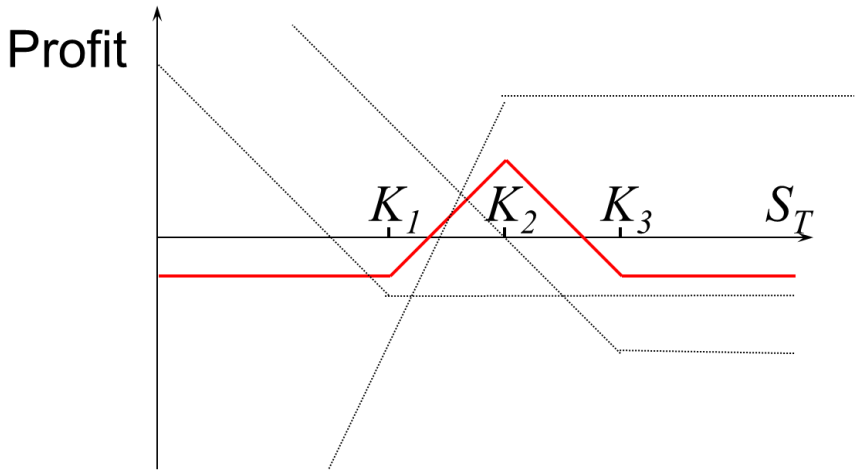
- Buy call with K_1
- Sell **two** calls with K_2
- Buy call with K_3

With *puts* has initial *outflow* :

- Buy put with K_1
- Sell **two** puts with K_2
- Buy put with K_3



Butterfly spread w/ calls



Butterfly spread w/ puts

Butterfly Spread Using Calls
Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*			
		$If\ S_T \le K_1$ (no exercise)	$If\ K_1 \le S_T \le K_2$ (one exercise)	$If\ K_2 \le S_T \le K_3$ (two exercise)	$If\ S_T \ge K_3$ (all exercise)
Long call w/ K_1	c_1	0	$S_T - K_1$	$S_T - K_1$	$S_T - K_1$
Short 2 calls w/ K_2	$-2c_2$	0	0	$2K_2 - 2S_T$	$2K_2 - 2S_T$
Long call w/ K_3	c_3	0	0	0	$S_T - K_3$
Total	$\boldsymbol{c_1 - 2c_2 + c_3}$	**0**	$\boldsymbol{S_T - K_1}$	$\boldsymbol{2K_2 - K_1 - S_T}$	**0**
	Total Profit:	$-(c_1 - 2c_2 + c_3)$	$S_T - K_1 - (c_1 - 2c_2 + c_3)$	$K_3 - S_T - (c_1 - 2c_2 + c_3)$	$-(c_1 - 2c_2 + c_3)$

Butterfly Spread Using Calls
Payoffs and Profit

There are two break-even spot prices:

$$S_T^{Break-even\,1} = K_1 + (c_1 - 2c_2 + c_3)$$

$$S_T^{Break-even\,2} = K_3 - (c_1 - 2c_2 + c_3)$$

$$Profit_T^{Minimum} = -(c_1 - 2c_2 + c_3)$$

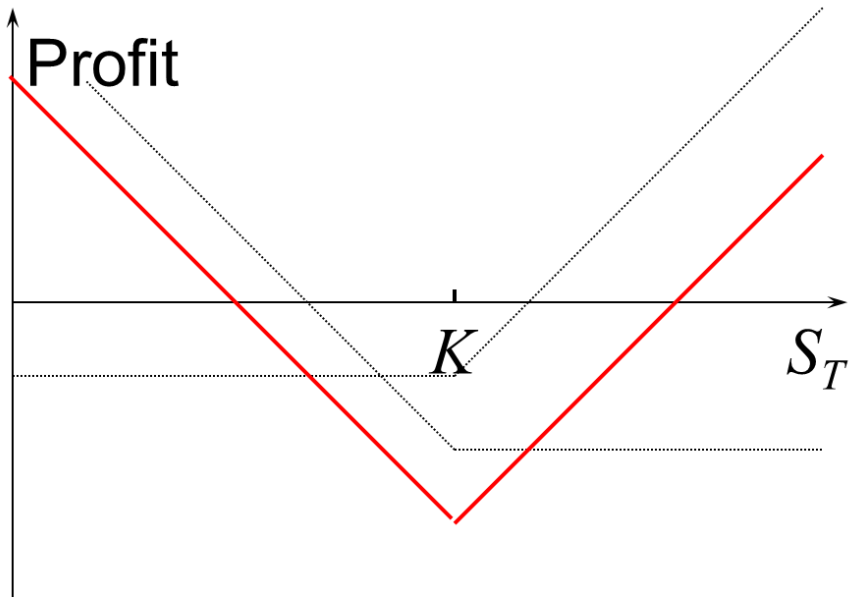
$$Profit_T^{Maximum} = K_2 - K_1 - (c_1 - 2c_2 + c_3) = K_3 - K_2 - (c_1 - 2c_2 + c_3)$$

Straddle Combination

- A spread strategy with a call and a put on the same underlying with same maturity and same strike
- Investor is hoping spot price will <u>either increase or decrease substantially</u>
- Limits downside risk, unlimited upside potential
- Bets on volatility, not price direction

Has an initial <u>outflow</u>:

- Buy call with K
- Buy put with K



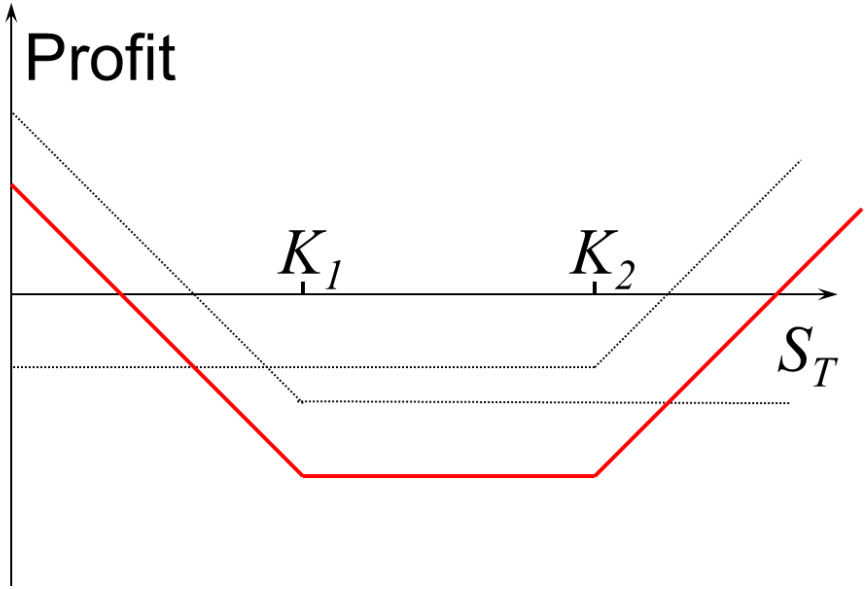
Straddle Combination Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*	
		$\boldsymbol{If\ S_T < K}$ **(exercise put)**	$\boldsymbol{If\ S_T \geq K}$ **(exercise call)**
Long call	c	0	$S_T - K$
Long put	p	$K - S_T$	0
Total	$\boldsymbol{c + p}$	$\boldsymbol{K - S_T}$	$\boldsymbol{S_T - K}$
	Total Profit:	$\boldsymbol{K - S_T - (c + p)}$	$\boldsymbol{S_T - K - (c + p)}$

- $\boldsymbol{S_T^{Break-even\ 1} = K - (c + p)}$
- $\boldsymbol{S_T^{Break-even\ 2} = K + (c + p)}$
- $\boldsymbol{Profit_T^{Minimum} = -(c + p)}$
- $\boldsymbol{Profit_T^{Maximum} = unlimited}$

Strangle Combination

- A spread strategy with a call and a put on the same underlying with same maturity but different strikes (call K_2>put K_1)
- Investor is hoping spot price will <u>either increase or decrease substantially</u>
  - Spot price change that leads to profits needs to be larger than with the straddle, but
  - Limits downside risk more than the straddle
- Has an initial <u>outflow</u>:
  - Buy put with K_1
  - Buy call with K_2



Cheaper than straddle, as here both options out of the money (in straddle at the money)

Strangle Combination Payoffs and Profit

Position	Initial *Cost*	Terminal *Payoff*		
		$If\ S_T \leq K_1$ (exercise put)	$If\ K_1 \leq S_T \leq K_2$ (no exercise)	$If\ S_T \geq K_2$ (exercise call)
Long put w/ K_1	p	$K_1 - S_T$	0	0
Long call w/ K_2	c	0	0	$S_T - K_2$
Total	$\boldsymbol{c + p}$	$\boldsymbol{K_1 - S_T}$	$\boldsymbol{0}$	$\boldsymbol{S_T - K_2}$
	Total Profit:	$\boldsymbol{K_1 - S_T - (c + p)}$	$\boldsymbol{-(c + p)}$	$\boldsymbol{S_T - K_2 - (c + p)}$

- $S_T^{Break-even\ 1} = K_1 - (c + p)$
- $S_T^{Break-even\ 2} = K_2 + (c + p)$
- $Profit_T^{Minimum} = -(c + p)$
- $Profit_T^{Maximum} = unlimited$

Other Payoff Patterns

- **Calendar spread** – similar to butterfly
  - Involves two calls (or two puts) with the same strike but different maturities
  - Buy longer maturity, sell shorter
  - Benefits from price stability, longer maturity option retains value

- **Strips & Straps** – similar to straddle; same strike and expiration
  - Strip – buy one call and two puts – decrease in spot more profitable
  - Strap – buy two calls and one put – increase in spot more profitable

- If options with all strike prices were available, any payoff pattern could (at least approximately) be created

Let’s practice!

1. Suppose that put options on a stock with strike prices \$30 and \$35 cost \$4 and \$7, respectively. How can the options be used to create (a) a bull spread and (b) a bear spread? Construct a table that shows the profit and payoff for both spreads.
2. A call with a strike price of \$60 costs \$6. A put with the same strike price and expiration date costs \$4. Construct a table that shows the profit from a straddle. For what range of stock prices would the straddle lead to a loss?
3. A trader sells a strangle by selling a call option with a strike price of \$50 for \$3 and selling a put option with a strike price of \$40 for \$4. For what range of prices of the underlying asset does the trader make a profit?
4. Three put options on a stock have the same expiration date and strike prices of \$55, \$60, and \$65. The market prices are \$3, \$5, and \$8, respectively. Explain how a butterfly spread can be created. Construct a table showing the profit from the strategy. For what range of stock prices would the butterfly spread lead to a loss?